

Year 11

Mathematics

Yearly Examination

2011

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your name and “N/A” on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Total Marks – 84
Attempt Questions 1–7
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)	Marks
(a) Evaluate $\sqrt{\frac{86.54 \times 3.16}{2.6^3}}$ correct to 2 significant figures.	2
(b) Express $\frac{3\sqrt{3}}{2\sqrt{3}-1}$ with a rational denominator.	2
(c) Factorise completely	
i) $x^2 - 25$	1
ii) $4x^2 - 8x + 3$	1
(d) Solve $ x - 8 = 6$	2
(e) Simplify $\frac{(3xy^3)^3}{3x^2y^4}$	2
(f) Convert $0.\dot{2}\dot{8}$ to a fraction.	2

Question 2 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ 2

(b) Consider the function

$$y = x^2 - 2x - 3$$

i) Find its y and x intercepts. 2

ii) Find the axis of symmetry and vertex. 2

iii) Sketch this function showing all important features. 2

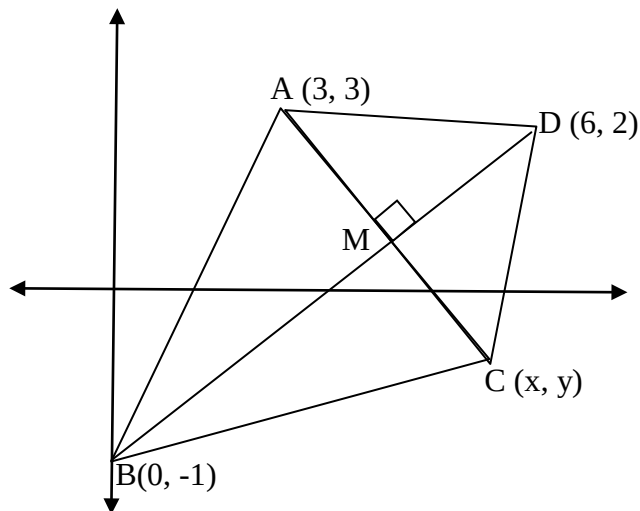
iv) State its Domain and Range. 2

(c) Show that $f(x) = x^3 - 25x$ is an odd function. 2

Question 3 (12 marks) Use a SEPARATE writing booklet

Marks

In the quadrilateral $ABCD$, the points A , B and D are $(3, 3)$, $(0, -1)$ and $(6, 2)$ respectively. The line BD bisects the line AC at right angles at the point M .



NOT TO SCALE

- | | | |
|-----|---|---|
| (a) | Find the exact distance BD . | 1 |
| (b) | Show that the gradient of BD is $\frac{1}{2}$. | 1 |
| (c) | Show that the equation of the line BD is $x - 2y - 2 = 0$. | 2 |
| (d) | Show that the equation of AC is $2x + y - 9 = 0$. | 2 |
| (e) | Show that the coordinates of M are $(4, 1)$ | 1 |
| (f) | Show that the coordinates of C are $(5, -1)$ | 2 |
| (g) | Find the area of the quadrilateral $ABCD$. | 3 |

Question 4 (12 marks) Use a SEPARATE writing booklet

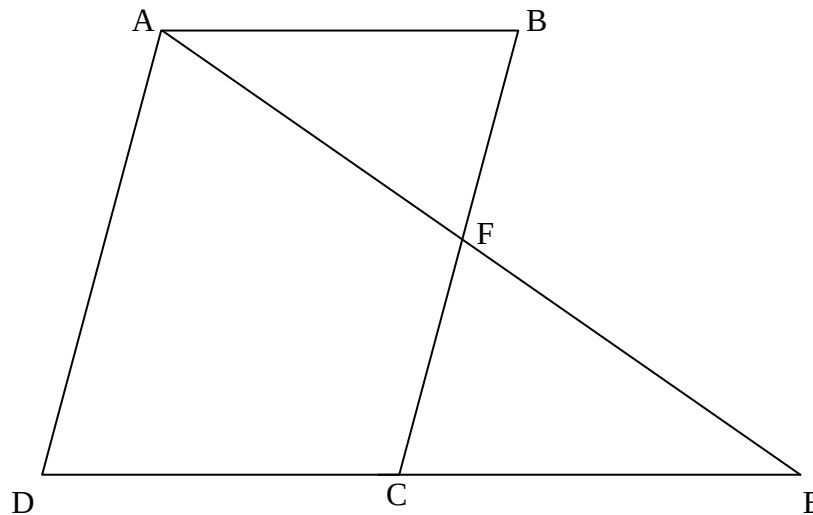
Marks

- (a) Find the acute angle which the line $2x - y + 3 = 0$ makes with the x axis to the nearest degree. 2

- (b) i) Find the perpendicular distance from the line $3x - 4y + 10 = 0$ to the point $(0,0)$ 2

- ii) Hence explain why the line $3x - 4y + 10 = 0$ is a tangent to the circle $x^2 + y^2 = 4$ 2

- (c) ABCD is a parallelogram. DC is produced to E. AE cuts BC at F. $AD=16\text{cm}$, $CE=9\text{cm}$ and $BF=10\text{cm}$.



- i) Prove that $\triangle ABF$ is similar to $\triangle ECF$. 2
- ii) Find AB. 2
- (d) Using the method of first principles, find $f'(x)$ if $f(x) = 2x^2 - 1$ 2

Question 5 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Differentiate each of the following functions.

i) $3x^4 - 2x + 1.$ 1

ii) $(2x^2 - 3)^4$ 1

iii) $(2x + 1)(5x^2 - 1)$ 2

iv) $\frac{2x}{x+3}$ 2

(b) For the function $y = 3x^2 + 4$ find the equation of the normal at $x = 2$. 3

(c) Find the equation of the line that makes an angle of 60° in the positive direction with the x axis and passes through the point (2, -1) 3

Question 6 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Find the exact value of

$$\cot 45^\circ + \sec 45^\circ$$

1

(b) Solve each of the following trigonometric equations.

i. $2 \sin \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$

2

ii. $\tan 2\theta = -1$ for $0^\circ \leq \theta \leq 360^\circ$

3

iii. $\cos \frac{\theta}{2} = \frac{1}{\sqrt{2}}$ for $-180^\circ \leq \theta \leq 180^\circ$

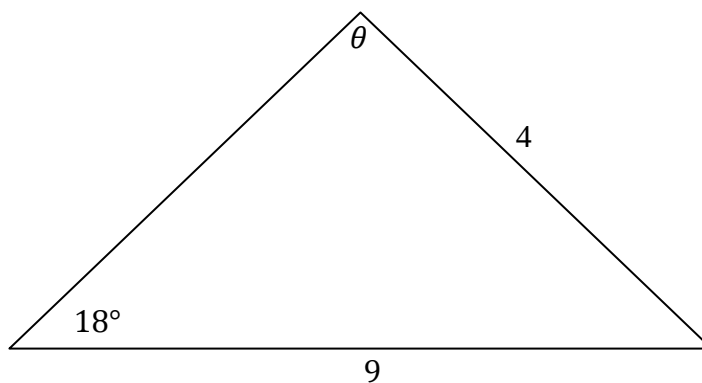
2

iv. $2 \sin^2 \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$

2

(c) Find the Value of θ (correct to the nearest degree)

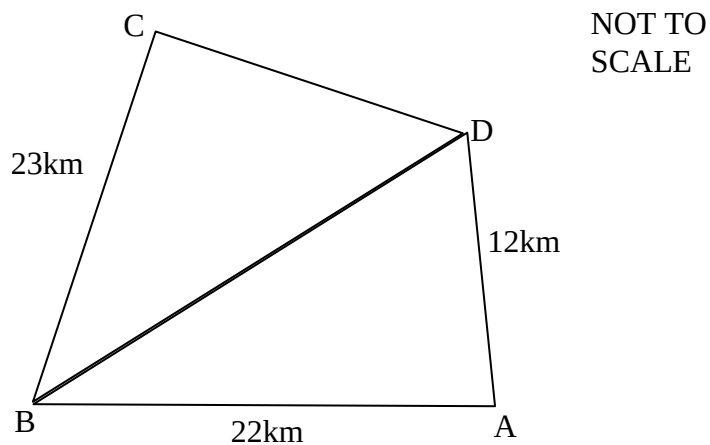
2



Question 7 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) If $\sin\theta = -\frac{4}{11}$ and $\tan\theta > 0$, find the exact value of $\cos\theta$. 2
- (b) Prove that $3\cos^2\theta - 1 = 2 - 3\sin^2\theta$ 3
- (c) The diagram represents a cattle station ABCD in the shape of a quadrilateral. A is 22 km due east of B and D is 12 km from A at a bearing of 030° . C is 23 km from B at a bearing of 030° and at a bearing of 300° from D.



- i) Copy or trace the diagram into your writing booklet and mark the lengths and the bearings on it. 1
- ii) Explain why $\angle BAD = 120^\circ$ and $\angle BCD$ is a right angle. 2
- iii) Find the length of BD, correct to one decimal place. 2
- iv) Find the area of the cattle station correct to the nearest square km. 2

End of Examination

2011 2U AT3 Prelim

$$1a) \sqrt{\frac{86,54 \times 3,16}{2 \cdot 6^3}} = 3,9445... \checkmark$$

$$= 3,9 \text{ 2SF } \checkmark$$

$$b) \frac{3\sqrt{3}}{2\sqrt{3}-1} = \frac{3\sqrt{3} \times (2\sqrt{3}+1)}{(2\sqrt{3}-1)(2\sqrt{3}+1)} \checkmark$$

$$= \frac{6 \times 3 + 3\sqrt{3}}{4 \times 3 - 1}$$

$$= \frac{18 + 3\sqrt{3}}{11} \checkmark$$

$$c) i) x^2 - 25 = (x+5)(x-5) \checkmark$$

$$ii) 4x^2 - 8x + 3$$

$$= 4x^2 - 2x - 6x + 3$$

$$= 2x(2x-1) - 3(2x-1)$$

$$= (2x-1)(2x-3) \checkmark$$

P: 12
S: -8
F: -2, -6.

$$d) |x-8| = 6$$

$$x-8=6 \quad -(x-8)=6$$

$$x=14 \quad -x+8=6$$

$$\quad \quad \quad -x=-2$$

$$\quad \quad \quad x=2$$

$$\therefore x = 2, 14 \quad \checkmark \checkmark$$

$$e) \frac{(3xy^3)^3}{3x^2y^4} = \frac{27x^3y^9}{3x^2y^4} \checkmark$$

$$= 9xy^5 \checkmark$$

$$f) x = 0, \dot{2} \dot{8}$$

$$= 0,282828...$$

$$100x = 28,2828...$$

$$99x = 28$$

$$x = \frac{28}{99} \quad \checkmark \checkmark$$

$$2a) \lim_{x \rightarrow 3} \frac{x^2-9}{x-3}$$

$$= \lim_{x \rightarrow 3} \frac{(x+3)(x-3)}{(x-3)} \checkmark$$

$$= \lim_{x \rightarrow 3} x+3 \quad \checkmark$$

$$= 6. \quad \checkmark$$

$$b) y = x^2 - 2x - 3$$

$$= (x-3)(x+1).$$

$$i) y \text{ int: } y = -3 \quad \checkmark$$

$$x \text{ int: } x = -1, 3 \quad \checkmark$$

$$ii) \text{ axis sym:}$$

$$x = -\frac{b}{2a} = \frac{2}{2}$$

$$= 1.$$

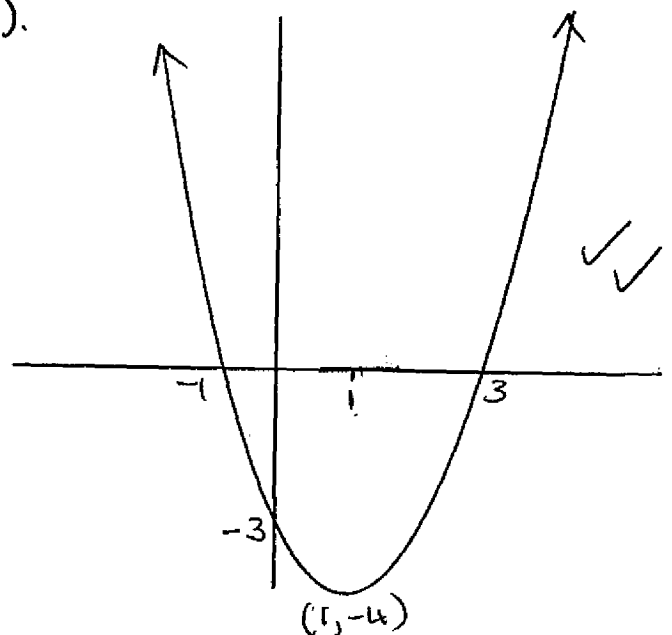
vertex:

$$y = 1 - 2 - 3 = -4$$

$$\therefore \text{axis: } x = 1 \quad \checkmark$$

$$\text{vertex: } (1, -4) \quad \checkmark$$

iii).



$$iv) D: \text{all } x \in \mathbb{R}. \quad \checkmark$$

$$R: y \geq -4. \quad \checkmark$$

g) If odd then $f(-x) = -f(x)$.

$$f(x) = x^3 - 25x$$

$$f(-x) = (-x)^3 - 25(-x)$$

$$= -x^3 + 25x$$

$$= -(x^3 - 25x)$$

$$= -f(x)$$

$\therefore$ odd. since $f(x) = -f(x)$

Q3.

$$A(3,3) \quad C(x,y)$$

$$B(0,-1) \quad D(6,2).$$

$$a) BD = \sqrt{6^2 + 3^2}$$

$$= \sqrt{45} = 3\sqrt{5} \text{ u.}$$

$$b) m_{BD} = \frac{2 - (-1)}{6 - 0}$$

$$= \frac{3}{6} = \frac{1}{2}$$

$$c) BD: y + 1 = \frac{1}{2}(x - 0)$$

$$y + 1 = \frac{1}{2}x$$

$$2y + 2 = x$$

$$x - 2y - 2 = 0$$

$$d) AC: AC \perp BD$$

$$\text{so } m_{AC} = -2.$$

$$y - 3 = -2(x - 3)$$

$$= -2x + 6$$

$$2x + y - 9 = 0$$

e) M is point of intersection
BD and AC.

$$x - 2y - 2 = 0 \quad \text{--- (1)}$$

$$2x + y - 9 = 0 \quad \text{--- (2)}$$

$$\textcircled{1} \times 2.$$

$$2x - 4y - 4 = 0 \quad \textcircled{3}.$$

$$\textcircled{2} - \textcircled{3}$$

$$5y - 5 = 0$$

$$y = 1$$

$$x - 2 - 2 = 0 \quad x = 4 \quad \therefore M(4,1)$$

f) M is midpt AC. $C(x,y)$

$$\text{so } 4 = \frac{3+x}{2} \quad 1 = \frac{3+y}{2} \quad \checkmark$$

$$8 = 3 + x \quad 2 = 3 + y$$

$$x = 5 \quad y = -1$$

$$\therefore C(5, -1)$$

$$g) \text{Area} = 2 \times \text{area } \triangle BAD$$

$$= 2 \times \frac{1}{2} \times BD \times AM$$

$$AM = \sqrt{(4-3)^2 + (1-3)^2}$$

$$= \sqrt{1+4}$$

$$= \sqrt{5} \text{ u}$$

$$\text{so } \text{Area} = BD \times AM$$

$$= 3\sqrt{5} \times \sqrt{5}$$

$$= 15 \text{ u}^2.$$

4.

$$a) 2x - y + 3 = 0$$

$$y = 2x + 3$$

$$m = 2.$$

$$\therefore \tan \theta = 2$$

$$\theta = 63^\circ 26'$$

$$= 63^\circ$$

$$b) 3x - 4y + 10 = 0$$

$$(0,0)$$

$$d = \left| \frac{0-0+10}{\sqrt{3^2+4^2}} \right|$$

$$= \frac{10}{5}$$

$$= 2 \text{ u.}$$

1) $x^2 + y^2 = 4$ is a circle

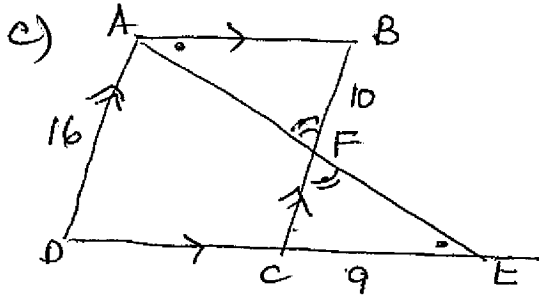
centre $(0,0)$ $r=2$

Since the $\perp$ distance b/w

$3x - 4y + 10 = 0$ and the

centre $(0,0)$ is = radius,

the line must be a tangent



1) $\triangle ABF \quad \triangle ECF$

$\angle BAF = \angle CEF$ alternate angles $AB \parallel CE$

$\angle AFB = \angle EFC$ vertically opposite angles

$\therefore \triangle ABF \parallel \triangle ECF$ equiangular

ii) $\frac{AB}{CE} = \frac{BF}{CF}$ corresponding sides similar
 Δ s same proportion

$$\frac{AB}{9} = \frac{10}{6}$$

$$AB = \frac{90}{6}$$

$$= 15$$

iii) $f(x) = 2x^2 - 1$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x+h)^2 - 1 - (2x^2 - 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 1 - 2x^2 + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2h(2x + h)}{h}$$

$$= \lim_{h \rightarrow 0} 4x + h$$

$$= 4x.$$

5. a)

i) $y = 3x^4 - 2x + 1$

$$y' = 12x^3 - 2$$

ii) $y = (2x^2 - 3)^4$

$$y' = 4 \cdot 4x(2x^2 - 3)^3$$

$$= 16x(2x^2 - 3)^3$$

iii) $y = (2x+1)(5x^2-1)$

$$u = 2x+1 \quad v = 5x^2-1$$

$$u' = 2 \quad v' = 10x$$

$$y' = 10x(2x+1) + 2(5x^2-1)$$

$$= 20x^2 + 10x + 10x^2 - 2$$

$$= 30x^2 + 10x - 2$$

iv) $y = \frac{2x}{x+3}$

$$u = 2x \quad v = x+3$$

$$u' = 2 \quad v' = 1$$

$$y' = \frac{2(x+3) - 2x}{(x+3)^2}$$

$$= \frac{2x+6-2x}{(x+3)^2}$$

$$= \frac{6}{(x+3)^2}$$

$$b) y = 3x^2 + 4.$$

$$y' = 6x.$$

$$\text{at } x=2 \quad m_T = 12. \quad m_N = -\frac{1}{12}$$

$$x=2 \quad y = 3(4) + 4 = 16.$$

$$\text{so } m_N = -\frac{1}{12} (2, 16)$$

$$y - 16 = -\frac{1}{12}(x - 2) \quad \checkmark$$

$$12y - 192 = -x + 2.$$

$$x + 12y - 194 = 0. \quad \checkmark$$

$$c) \tan 60^\circ = m.$$

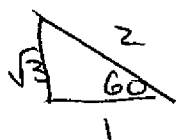
$$\therefore m = \sqrt{3}. \quad \checkmark$$

$$(2, -1).$$

$$y + 1 = \sqrt{3}(x - 2) \quad \checkmark$$

$$= \sqrt{3}x - 2\sqrt{3}$$

$$\sqrt{3}x - y - 2\sqrt{3} - 1 = 0. \quad \checkmark$$



$$6a) \cot 45^\circ + \sec 45^\circ$$

$$= 1 + \sqrt{2}$$

$$= 1 + \sqrt{2}. \quad \checkmark$$

$$b) i) 2\sin\theta = 1 \quad 0^\circ \leq \theta \leq 360^\circ$$

$$\sin\theta = \frac{1}{2} \quad Q1, Q2, 30^\circ$$

$$\theta = 30^\circ, 150^\circ \quad \checkmark \checkmark$$

$$ii) \tan 2\theta = -1 \quad 0^\circ \leq \theta \leq 360^\circ$$

$$0^\circ \leq 2\theta \leq 720^\circ$$

$$Q2, Q4, 45^\circ$$

$$2\theta = 135^\circ, 315^\circ$$

$$495^\circ, 675^\circ \quad \checkmark$$

$$\theta = 67.5^\circ, 157.5^\circ,$$

$$247.5^\circ, 337.5^\circ \quad \checkmark \checkmark$$

$$iii) \cos \frac{\theta}{2} = \frac{1}{\sqrt{2}} \quad -180^\circ \leq \theta \leq 180^\circ$$

$$-90^\circ \leq \frac{\theta}{2} \leq 90^\circ$$

$$Q1, Q4, 45^\circ.$$

$$-Q1, -Q4.$$

$$\frac{\theta}{2} = -\cancel{315^\circ} \quad -Q1 \quad Q1 \quad Q4$$

$$-45^\circ \quad 45^\circ \quad 315^\circ$$

not in domain $\rightarrow$

$$\theta = -90^\circ, 90^\circ. \quad \checkmark \checkmark$$

$$iv) 2\sin^2\theta = 1 \quad 0^\circ \leq \theta \leq 360^\circ$$

$$\sin^2\theta = \frac{1}{2}$$

$$\sin\theta = \pm \frac{1}{\sqrt{2}} \quad \text{all } Q, 45^\circ$$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ.$$

$$c) \frac{\sin\theta}{9} = \frac{\sin 18^\circ}{4}$$

$$\sin\theta = \frac{9\sin 18^\circ}{4} \quad \checkmark$$

$$= 0.6952 \dots$$

$$\theta = 44^\circ 3'$$

$$= 44^\circ \quad \checkmark$$

$$7(a) \sin\theta = -\frac{4}{11} \quad \tan\theta > 0$$

$$\therefore Q3.$$



$$\cos\theta = -\frac{\sqrt{105}}{11} \quad \checkmark \checkmark$$

$$b) 3\cos^2\theta - 1 = 2 \Rightarrow 3\sin^2\theta$$

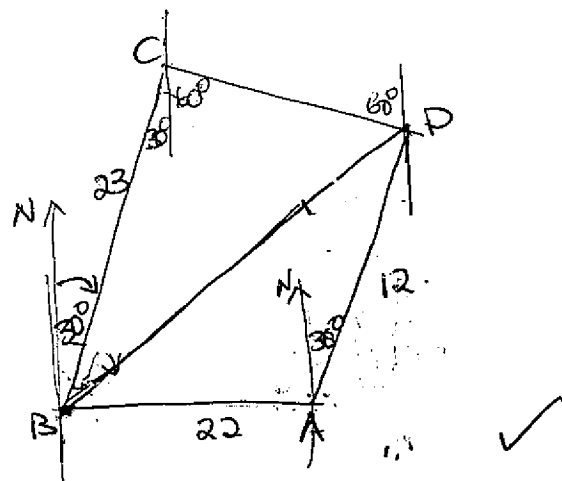
$$\text{LHS} = 3(1 - \sin^2\theta) - 1 \quad \checkmark$$

$$= 3 - 3\sin^2\theta - 1$$

$$= 2 - 3\sin^2\theta$$

$$= \text{RHS} \quad \checkmark$$

1)


$$\therefore \angle NPA = 90^\circ$$

$$\begin{aligned}\text{So } \angle BAO &= 90^\circ + 30^\circ \\ &= 120^\circ\end{aligned}$$

$$\begin{aligned}\angle BCD &= 30^\circ + 60^\circ \\ &= 90^\circ\end{aligned}$$

$$u) \quad BD^2 = 12^2 + 22^2 - 2 \times 12 \times 22 \times \cos 120^\circ$$
$$= 892$$

$$BD = 29.866 \dots$$

$$= 29.9 \text{ km.}$$

iv) Area = Area $\Delta BCD +$

$$CD = \sqrt{BD^2 - BC^2} = \sqrt{892 - 23^2} = 19.05$$
$$\Delta BCD = \frac{1}{2} \times 23 \times 19.05$$

$$= 219.075 \text{ km}^2$$

$$\Delta BDA = \frac{1}{2} \times 22 \times 12 \times \sin 120^\circ$$
$$= 114.32 \text{ km}^2 \quad \checkmark$$

Total = 333.395..

$$= 333 \text{ km}^2$$